

Pradyuman

Software Engineer (Backend & Systems) — Go, Rust — Distributed Systems & Storage

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SUMMARY

Senior Backend and Systems Engineer (4+ YOE) with deep hands-on experience in distributed systems, storage, and observability. Strong background in Go and Rust with a focus on performance, reliability, and production-grade system design.

SKILLS

Languages & Systems: Go, Rust, C++, Python • Linux internals, concurrency, networking

Distributed Systems: Storage and observability systems, partitioning & replication concepts

Infrastructure: Kubernetes, Docker, Terraform, GitHub Actions • OpenTelemetry, Prometheus, Grafana • AWS, GCP, Azure

EXPERIENCE

CtrlB

April 2024 – Present

Senior Software Engineer → Lead Software Engineer

Bengaluru, Karnataka

- Core contributor to a high-performance data engine written in Rust, owning critical query execution paths and performance optimizations; reduced end-to-end query latency by 30% and significantly lowered crash frequency under sustained peak load by addressing memory and concurrency bottlenecks.
- Designed and owned backend services in Go for authentication, RBAC, and multi-tenant isolation, defining clear trust boundaries and enforcing tenant-level data separation for a unified lake architecture.
- Built the observability control plane responsible for lifecycle management of OpenTelemetry agents, enabling centralized configuration, upgrades, and policy enforcement across 100+ nodes in production environments used by multiple companies, while reducing configuration drift and operational toil.
- Provisioned a self-hosted Kubernetes cluster from first principles, defining networking, storage, and security baselines; reduced onboarding time from 4–5 hours to under 15 minutes through fully automated setup.

Middleware

Sept 2023 – Feb 2024

Senior Software Engineer

Ahmedabad, Gujarat

- Designed modular, reusable deployment workflows using Helm and Kubernetes manifests, enabling rapid and consistent agent provisioning across Kubernetes clusters.
- Built observability integrations for ClickHouse, Redpanda, and Cassandra, enabling teams to monitor performance bottlenecks, resource utilization, and failure modes in production systems.
- Automated CI/CD pipelines using GitHub Actions, improving release reliability and reducing manual deployment effort by 40%.

Gap Inc.

June 2021 – Sept 2023

Software Engineer

Hyderabad, Telangana

- Led OpenTelemetry adoption across a large AKS-based microservices ecosystem (100+ services), building a vendor-neutral observability layer to standardize metrics and logs.
- Led a production OS migration from RHEL to Oracle Enterprise Linux across critical workloads, achieving \$50K annual cost savings while maintaining SLA-grade availability and performance.
- Built infrastructure-as-code with Chef to standardize and enforce VM provisioning, improving security posture and configuration consistency across environments.

PROJECTS

Mini-NFS Protocol Implementation | Rust | 🐙

Systems Project

- Built a custom NFS-like protocol over TCP to explore low-level networking and file transfer mechanisms, with emphasis on concurrency control and protocol correctness.
- Implemented file server and client components in Rust using line-based command parsing and streaming byte-level I/O for reliable remote file access.

Mini Object Store (S3-inspired) | Golang | 🐙

Storage Systems Project

- Designed and partially implemented a minimal object storage service inspired by S3 to study metadata layout, indexing strategies, and request routing in storage systems.
- Implemented core object operations (PUT, GET, LIST) with a simple on-disk layout and metadata indexing to support efficient object lookup.

EDUCATION

National Institute of Technology (NIT), Hamirpur

Bachelor of Technology in Computer Science

2017 – 2021

Hamirpur, HP